

Chapter 105

I/O Nodes

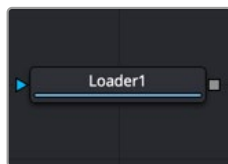
This chapter details the input and output of media using Loader and Saver nodes within Fusion Studio as well as the MediaIn and MediaOut nodes in DaVinci Resolve.

The abbreviations next to each node name can be used in the Select Tool dialog when searching for tools and in scripting references.

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Loader Node [Ld]



The Loader node

Loader Node Introduction

NOTE: The Loader node in DaVinci Resolve is only used for importing EXR files.

When using Fusion Studio, the Loader node is the node you use to select and load footage from your hard drive into the Node Editor. There are three ways to add a Loader node, and consequently a clip, to the Node Editor.

- Add the Loader from the Effects Library or toolbar (Fusion Studio only), and then use Loader's file browser to bring a clip into the Node Editor
- Drag clips from an OS window directly into the Node Editor, creating a Loader node in the Node Editor.
- Choose File > Import > Footage (Fusion Studio only), although this method creates a new composition as well as adds the Loader node to the Node Editor.

When a Loader is added to the Node editor, a File dialog is displayed automatically to allow the selection of a clip from your hard drives.

NOTE: You can disable the automatic display of the file browser by disabling Auto Clip Browse in the Global > General Preferences.

Once clips are brought in using the Loader node, the Loader is used for trimming, looping, and extending the footage, as well as setting the field order, pixel aspect, and color depth. The Loader is arguably the most important tool in Fusion Studio.

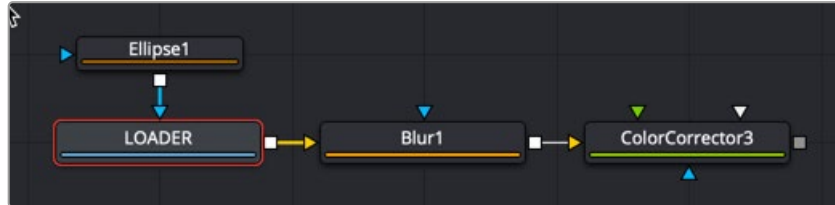
Inputs

The single input on the Loader node is for an effect mask to crop the image brought in by the Loader.

Effect Mask: The blue input is for a mask shape created by polylines, basic primitive shapes, paint strokes, or bitmaps from other tools. Connecting a mask to this input limits the loaded image to appear only within the mask. An effects mask is applied to the tool after the tool is processed.

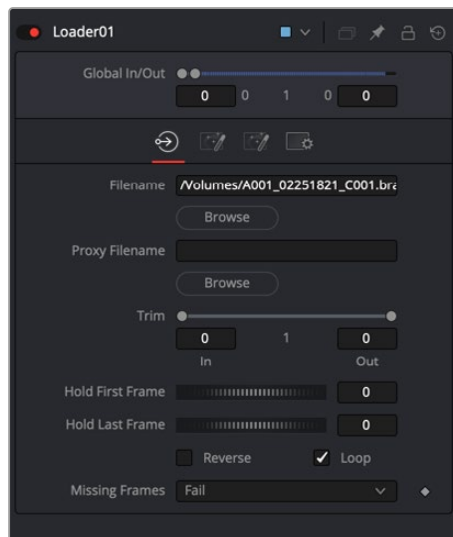
Basic Node Setup

The Loader node is a 2D image of any format supported in Fusion Studio. It is limited to an EXR format in DaVinci Resolve. Below, the LOADER imports an image, which is then masked using an Ellipse matte. The output of the masked LOADER is passed onto 2D image-processing nodes.



A Loader node used for importing images

Inspector



Loader node File tab

File Tab

The File tab for the Loader includes controls for trimming, creating a freeze frame, looping, and reversing the clip. You can also reselect the clip that the Loader links to on your hard drive.

Global In and Out

The Global In and Out handles are used to specify the position of this node within the project. Use Global In to specify the frame on which that the clip starts and use Global Out to specify the frame on which the clip ends within the project's global range. The node does not produce an image on frames outside of this range.

If the Global In and Out values are decreased to the point where the range between the In and Out values is smaller than the number of available frames in the clip, Fusion automatically trims the clip by adjusting the Trim range control. If the Global In/Out values are increased to the point where the range between the In and Out values is larger than the number of available frames in the clip, Fusion automatically lengthens the clip by adjusting the Hold First/Last Frame controls. Extended frames are visually represented in the range control by changing the color of the held frames to green in the control.

To slide the clip in time or move it through the project without changing its length, place the mouse pointer in the middle of the range control and drag it to the new location, or enter the value manually in the Global In value box.

Filename

The Filename field shows the file path of the clip imported to the Node Editor by the Loader node. Clicking on the Browse button opens a standard file browser. The path to the footage can also be typed directly using the field provided. The text box supports filename completion. As the name of a directory or file is typed in the text box, Fusion displays a pop-up that lists possible matches. Use the arrow keys to select the correct match and complete the path.

NOTE: Loading image sequences is common practice for compositing, whether the image sequence comes from a 3D renderer or a digital cinema camera. If the last part of a file's name is a number (not counting the file extension), Fusion automatically scans the directory looking for files that match the sequence. For example, the following filenames would be valid sequences:

`image.0001.braw , image.0002.braw, image.0003.braw ...`

or

`image151.exr , image152.exr, image153.exr ...`

The following would not be considered a sequence since the last characters are not numeric.

`shot.1.fg.jpg, shot.2.fg.jpg, shot.3.fg.jpg`

It is not necessary to select the first file in the sequence. Fusion searches the entire folder for files matching the sequence in the selected filename. Also, Fusion determines the length of the sequence based on the first and last numeric value in the filenames. Missing frames are ignored. For example, if the folder contains two files with the following names:

`image.0001.exr, image.0100.exr`

Fusion sees this as a file sequence with 100 frames, not an image sequence containing two frames. The Missing Frames drop-down menu is used to choose how Fusion handles missing frames.

The Trim In/Trim Out control's context menu can also be used to force a specific clip length or to rescan the folder. Both controls are described in greater detail below.

Occasionally, you want to load only a single frame out of a sequence—e.g., a photograph from a folder containing many other files as well. By default, Fusion detects those as a sequence, but if you hold Shift while dragging the file from the OS window to the Node Editor, Fusion takes only that specific file and disregards any sequencing.

Proxy Filename

The Proxy Filename control only appears once the filename control points to a valid clip. The Proxy Filename can specify a clip that is loaded when the Proxy mode is enabled. This allows smaller versions of the image to be loaded to speed up file I/O from disk, and processing. For example, create a 1/4-scale version of an 8K EXR sequence to use as EXR proxy files. Whenever the Proxy mode of the Composition is enabled, the smaller resolution proxy clip is loaded from disk, and all processing is performed at the lower resolution, significantly improving render times. This is particularly useful when working with large RAW plates stored on a remote file server. Lower-resolution versions of the plates can be stored locally, reducing network bandwidth, interactive render times, and memory usage.

The proxy clip must have the same number of frames as the source clip, and the sequence numbers for the clip must start and end on the same frame numbers. It is strongly suggested that the proxies are the same format as the main files. In the case of formats with options, such as Cineon, DPX, and OpenEXR, the proxies use the same format options as the primary.

Trim

The Trim range control is used to trim frames from the start or end of a clip. Adjust the Trim In to remove frames from the start and adjust Trim Out to specify the last frame of the clip. The values used here are offsets. A value of 5 in Trim In would use the fifth frame in the sequence as the start, ignoring the first four frames. A value of 95 would stop loading frames after the 95th.

Hold First Frame/Hold Last Frame:

The Hold First Frame and Hold Last Frame controls hold the first or last frame of the clip for the specified amount of frames. Held frames are included in a loop if the footage is looped.

Reverse

Select this checkbox to reverse the footage so that the last frame is played first, and the first frame is played last.

Loop

Select this checkbox to loop the footage until the end of the project. Any lengthening of the clip using Hold First/Last Frame or shortening using Trim In/Out is included in the looped clip.

Missing Frames

The Missing Frames menu determines the Loader behavior when a frame is missing or is unable to load for any reason.

- **Fail:** The Loader does not output any image unless a frame becomes available. Rendering is aborted.
- **Hold Previous Output:** The last valid frame is held until a frame becomes available again. This fails if no valid frame has been seen—for example, if the first frame is missing.
- **Output Black:** Outputs a black frame until a valid frame becomes available again.
- **Wait:** Fusion waits for the frame to become available, checking every few seconds. This is useful for rendering a composition simultaneously with a 3D render. All rendering ceases until the frame appears.

The Magic Comp Variable

Loaders and Savers use the absolute file paths for the location of media. However, if you are using Fusion Studio, you can use a file path that is relative to the saved composition location. The Comp variable works for Loaders and Savers and helps you to keep your work organized. Entering Comp:\ in place of the full file path name is a shortcut for the folder where your Fusion composition document is saved.

You can either enter the Comp:\ manually into the filename field of a Loader, or turn on the Enable Reverse Mapping of Paths Preferences checkbox in the Path Map preferences. Enabling the Path Map preference check box will use the Comp:\ automatically.

So as long as all your source footage is stored in subfolders of your Comp folder, Fusion finds that footage regardless of the actual hard drive or network share name.

You could, for example, copy an entire shot from the network to your local drive, set up your Loaders and Savers to use the Comp variable, work all your magic locally (i.e., set up your composition), and then copy just the composition back to the server and issue a net-render. All render slaves automatically find the source footage.

Some examples:

Your composition is stored in:

X:\Project\Shot0815\Fusion\Shot0815.comp

Your source footage sits in:

X:\Project\Shot0815\Fusion\Greenscreen\0815Green_0000.dpx

The relative path in the Loader node would then be:

Comp:\Greenscreen\0815Green_0000.dpx

If your source footage is stored in:

X:\Project\Shot0815\Footage\Greenscreen\0815Green_0000.dpx

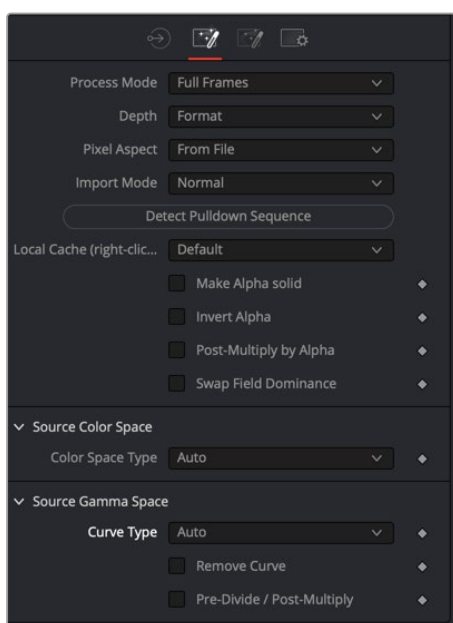
The relative path in the Loader node would then be:

Comp:\..\Footage\ Greenscreen\0815Green_0000.dpx

Observe how the two dots .. set the directory to go up one folder, much like CD .. in a Command Shell window.

Import Tab

The Import tab includes settings for the frame format and how to deal with fields, pixel aspect, 3:2 pull down/pull up conversion, and removing gamma curve types for achieving a linear workflow.



Loader Import tab

Process Mode

Use this menu to select the Fields Processing mode used by Fusion when loading the image. The Has Fields checkbox control in the Frame Format preferences determines the default option, and the default height as well. Available options include:

- Full frames
- NTSC fields
- PAL/HD fields
- PAL/HD fields (reversed)
- NTSC fields (reversed).

The two reversed options load fields in the opposite order and thus result in the fields being spatially swapped both in time order and in vertical order as well.

Use the Swap Field Dominance checkbox (described below) to swap fields in time only.

Depth

The Depth menu is used to select the color depth used to process footage from this Loader. The default option is Format.

- **Format:** The color depth is determined by the color depth supported in the file format loaded. For example, JPEG files automatically process at 8 bit because the JPEG file format does not store color depths greater than 8. EXR files load at Float. If the color depth of the format is undetermined, the default depth defined in the Frame Format preferences is used. Formats that support multiple color depths are set to the appropriate color depth automatically.
- **Default:** The color depth is determined by the settings in the composition's Frame Format Preferences panel.
- **Int 8 Bit/Int 16 Bit/Float 16/Float 32:** These options set the color depth for processing the image.

Pixel Aspect

This menu is used to determine the image's pixel aspect ratio.

- **From File:** The loader conforms to the image aspect detected in the saved file. There are a few formats that can store aspect information. TIFF, JPEG, and OpenEXR are examples of image formats that may have the pixel aspect embedded in the file's header. When no aspect ratio information is stored in the file, the default frame format method is used.
- **Default:** Any pixel aspect ratio information stored in the header of the image file is ignored. The pixel aspect set in the composition's frame format preferences is used instead.
- **Custom:** Select this option to override the preferences and set a pixel aspect for the clip manually. Selecting this button causes the X/Y Pixel Aspect control to appear.

Custom Pixel Aspect

This control is visible only when Custom is selected from the Pixel Aspect menu. Enter the desired X and Y aspect or right-click on the control to display a menu of common frame formats and their aspects.

Import Mode

This menu provides options for removing pull-up from an image sequence. Pull-up is a reversible method of combining frames used to convert 24 fps footage into 30 fps. It is commonly used to broadcast NTSC versions of films.

- **Normal:** This passes the image without applying pull-up or pull-down 2:3.
- **Pull Up:** This removes existing 3:2 pull-down applied to the image sequence, converting from 30 fps to 24 fps 2:3.
- **Pull Down:** The footage has pull-down applied, converting 24 fps footage to 30 fps by creating five frames out of every four. The process mode of a Loader set to Pull Down should always be Full Frames.

First Frame

This menu appears when the Import Mode is set to either Pull Up or Pull Down. It is used to determine which frame of the 3:2 sequence is used as the first frame of the loaded clip.

Detect Pull-Down Sequence

This button is used to detect and set the pull-up sequence of the footage automatically. It only works if Pull-Up or Pull-Down is first selected from the Import Mode menu. If it succeeds in detecting the order, the First Frame control automatically sets to the correct value.

Make Alpha Solid

When enabled, the original Alpha channel of the clip is cleared and set to solid white (completely opaque).

Invert Alpha

When enabled, the original Alpha channel of the clip is inverted. This may also be used in conjunction with Make Alpha Solid to set the Alpha to pure black (completely transparent).

Post-Multiply by Alpha

Enabling this option causes the color value of each pixel to be multiplied by the Alpha channel for that pixel. This option can be used to convert subtractive (non-premultiplied) images to additive (premultiplied) images.

Swap Field Dominance

When enabled, the field order (dominance) of the image is swapped, so the order in time the fields appear is reversed. Unlike the Process Mode control, this is done without spatially swapping the scanlines in the image.

Color Space Type

This menu is used to set the Color Space of the footage to help achieve a linear color space workflow. Unlike the Gamut tool, this doesn't perform any actual color space conversion, but instead adds the source color space data into the metadata, if that metadata doesn't already exist. The metadata can then be used downstream by a Gamut tool with the From Image option, or in a Saver if explicit output spaces are defined there.

- **Auto:** Passes along any metadata that might be in the incoming image.
- **Space:** Allows the user to set the color space based on the recording device used to capture content or software settings used when rendering the content in another application.

Curve Type

This menu is used to determine the gamma curve of the footage. Once the Gamma Curve Type is set, you can choose to remove the curve to help achieve a linear workflow.

- **Auto:** Passes along any metadata that might be in the incoming image.
- **Space:** Allows the user to set the gamma curve based on the recording device used to capture content or software settings used when rendering the content in another application.
- **Log:** Displays the Log/Lin settings, similar to the Cineon Log node.

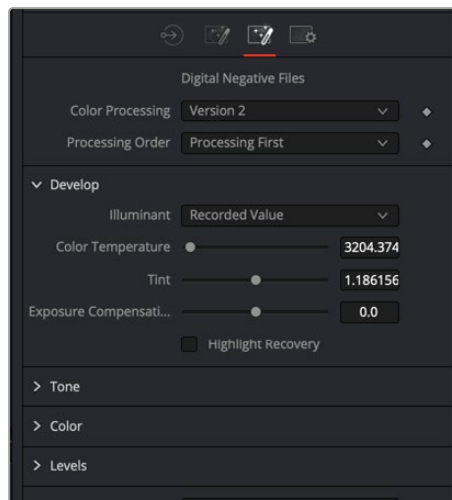
For more information on the Log settings, see *Chapter 99, "Film Nodes,"* in the DaVinci Resolve Reference Manual and Chapter 39, in the Fusion Reference Manual.

Remove Curve

Depending on the selected Curve Type or on the Gamma Space found in Auto mode, the associated Gamma Curve is removed from, or a log-lin conversion is performed on, the material, effectively converting it to a linear output space.

Format Tab

The Format tab contains file format-specific controls that dynamically change based on the selected Loader and the file it links to. Some formats contain a single control or no controls at all. Others like Camera RAW formats contain RAW-specific debayering controls. A partial format list is provided below for reference.



Format tab

- **OpenEXR:** EXR provides a compact and flexible format to support high dynamic range images (float). The format also supports a variety of extra channels and metadata. The Format tab for OpenEXR files provides a mechanism for mapping any non-RGBA channels to the channels supported natively in Fusion. Using the Format tab, you can enter the name of a channel contained in the OpenEXR file into any of the edit boxes next to the Fusion channel name. A command line utility for dumping the names of the channels can be found at <https://www.openexr.com>.
- **QuickTime:** QuickTime files can potentially contain multiple tracks. Use the format options to select one of the tracks.

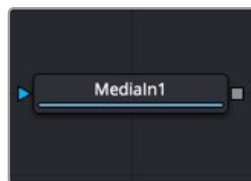
- **Cinema DNG:** CinemaDNG is an open format capable of high-resolution raw image data with a wide dynamic range. It was one of the formats recorded by Blackmagic Design cameras before switching over to the BRAW format.
- **Photoshop PSD Format:** Fusion can load any one of the individual layers stored in the PSD file, or the completed image with all layers. Transformation and adjustment layers are not supported. To load all layers in a PSD file with appropriate blend modes, use File > Import > PSD.

Common Controls

Settings Tab

The Settings tab controls are common to both Loader and Saver nodes, so their descriptions can be found in “The Common Controls” section at the end of this chapter.

MediaIn Node [MI]



The MediaIn node

MediaIn Node Introduction

NOTE: The MediaIn node is only available in DaVinci Resolve.

The MediaIn node is the foundation of every composition you create in DaVinci Resolve's Fusion page. In most cases, it replaces the Loader node used in Fusion Studio for importing clips. There are four ways to add a MediaIn node to the Node Editor.

- In the Edit or Cut page, position the playhead over a clip in the Timeline, and then click the Fusion page button. The clip from the Edit or Cut page Timeline is represented as a MediaIn node in the Node Editor.
- Drag clips from the Media Pool into the Node Editor, creating a MediaIn node in the Node Editor.
- Drag clips from an OS window directly into the Node Editor, creating a MediaIn node in the Node Editor.
- Choose Fusion > Import > PSD when importing PSD files into the Node Editor. Each PSD layer is imported as a separate MediaIn node.

NOTE: Although a MediaIn tool is located in the I/O section of the Effects Library, it is not used as a method to import clips.

When clips are brought in from the Media Pool, dragged from the OS window, or via the Import PSD menu option, you can use the MediaIn node's Inspector for trimming, looping, and extending the footage, as well as setting the source's color and gamma space.

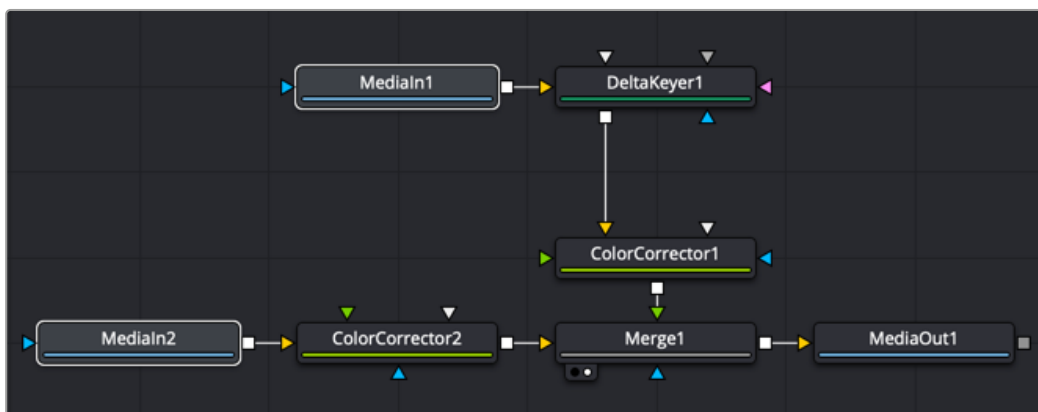
Inputs

The single input on the MediaIn node is for an effect mask to crop the image brought in by the MediaIn.

Effect Mask: The blue input is for a mask shape created by polylines, basic primitive shapes, paint strokes, or bitmaps from other tools. Connecting a mask to this input limits the source image to appear only within the mask. An effects mask is applied to the tool after the tool is processed.

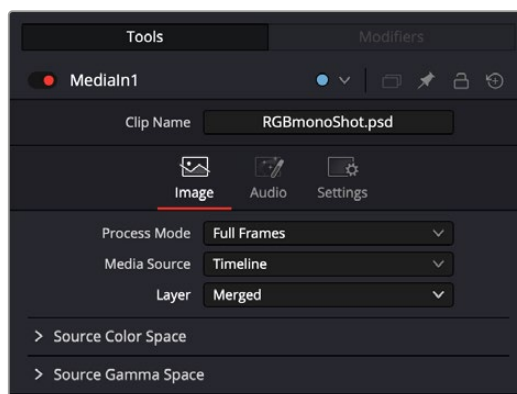
Basic Node Setup

The MediaIn node is typically the starting point for all composites done in the Fusion page of DaVinci Resolve. It contains the clip from the Edit page or Cut page. Any clip brought in from the Media Pool is also added as a MediaIn node.



Two MediaIn nodes: one from the Edit page Timeline and one from the Media Pool

Inspector



MediaIn node Image tab

Image Tab

When brought in from the Media Pool or dragged from an OS window, the MediaIn node's Image tab includes controls for trimming, creating a freeze frame, looping, and reversing the clip. You can also reselect the clip the MediaIn links to on your hard drive. A subset of these controls is available when the MediaIn node is brought in from the Edit or Cut page Timeline.

Global In and Out

Only used when a clip is brought in through the Media Pool or an OS window, the Global In and Out handles are used to specify the start and end of this node within the Fusion effect. Use Global In to specify the frame on which the clip starts and use Global Out to specify the frame on which the clip ends within the comp's global range. The node does not produce an image on frames outside of this range.

If the Global In and Out values are decreased to the point where the range between the In and Out values is smaller than the number of available frames in the clip, Fusion automatically trims the clip by adjusting the Trim range control. If the Global In/Out values are increased to the point where the range between the In and Out values is larger than the number of available frames in the clip, Fusion automatically lengthens the clip by adjusting the Hold First/Last Frame controls. Extended frames are visually represented in the range control by changing the color of the held frames to green in the control.

To slide the clip in time or move it through the project without changing its length, place the mouse pointer in the middle of the range control and drag it to the new location, or enter the value manually in the Global In value box.

- **Clip Name:** Displays the name of that clip.

Process Mode

Use this menu to select the Fields Processing mode used by Fusion when loading the image. The Has Fields checkbox control in the Frame Format preferences determines the default option, and the default height as well. Available options include:

- Full frames
- NTSC fields
- PAL/HD fields
- PAL/HD fields (reversed)
- NTSC fields (reversed).

The two reversed options load fields in the opposite order and thus result in the fields being spatially swapped both in time order and in vertical order as well.

Media Source: This control allows you to choose where media is sourced from; the three available options include: Timeline, Background, and Media Pool.

- **Timeline:** Links the clip from the Edit Timeline (default).
- **Background:** This pulls frames from the lower video tracks in the edit. When toggling the Media Source to Background, the composited result of all the lower tracks is visible.
- **Media Pool:** Links the clip directly from the Media Pool, bypassing the Edit Timeline.

Layer: In the case of Multilayer or Photoshop PSD files, this selects the layer in the file to use. For a Fusion Clip created using multiple video layers on the timeline, this will refer to the index number of each of the layers.

Trim

The Trim range control is used to trim frames from the start or end of a clip. Adjust the Trim In to remove frames from the start and adjust Trim Out to specify the last frame of the clip. The values used here are offsets. A value of 5 in Trim In would use the fifth frame in the sequence as the start, ignoring the first four frames. A value of 95 would stop loading frames after the 95th frame.

Hold First Frame/Hold Last Frame:

The Hold First Frame and Hold Last Frame controls hold the first or last frame of the clip for the specified amount of frames. Held frames are included in a loop if the footage is looped.

Reverse

Select this checkbox to reverse the footage so that the last frame is played first, and the first frame is played last.

Loop

Select this checkbox to loop the footage until the end of the project. Any lengthening of the clip using Hold First/Last Frame or shortening using Trim In/Out is included in the looped clip.

Source Color Space

Lets you choose a color space for the image data output by this Medialn node.

- **Auto:** uses the Timeline color space, or whichever color space is assigned by Resolve Color Management (RCM) if it's enabled.
- **Space:** Lets you choose a specific setting from a Color Space drop-down menu, while a visual “horseshoe” graph lets you see a representation of the color space you've selected.

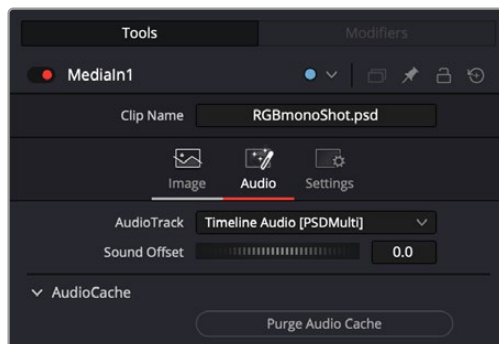
Source Gamma Space

Lets you choose a gamma setting for the image data output by this Medialn node. Once the gamma curve type is set, you can choose to remove the curve to help achieve a linear workflow.

- **Auto:** Uses the Timeline gamma, or whichever gamma is assigned by Resolve Color Management (RCM) if it's enabled.
- **Space:** Lets you choose a specific setting from a Gamma Space drop-down menu, while a visual graph lets you see a representation of the gamma setting you've selected.
- **Log:** Displays the Log Type drop-down menu where you can choose a specific log encoding profile. A visual graph shows a representation of the log setting chosen from the menu. Additional Lock RGB, Level, Soft Clip, Film Stock Gamma, Conversion Gamma, and Conversion table options are presented to finesse the gamma output.
- **Remove Curve:** The associated gamma curve is removed from, or a log-lin conversion is performed on, the material, effectively converting it to a linear output space.
- **Pre-Divide/Post-Multiply:** Lets you convert “straight” Alpha channels into premultiplied Alpha channels, when necessary.

Audio Tab

The Inspector for the Medialn node contains an Audio tab, where you can choose to solo the audio from the clip or hear all the audio tracks in the Timeline.



The Audio tab in the Medialn node is used to select the track for playback, slip the audio timing, and reset the audio cache.

If the audio is out of sync when playing back in Fusion, the Audio tab's Sound Offset wheel allows you to slip the audio in subframe frame increments. The slipped audio is only modified in the Fusion page. All other pages retain the original audio placement.

Audio with Media Pool Clips

Audio from a clip brought in through the Media Pool is muted by default. Hearing the audio from a Media Pool clip is a two step process.

To hear audio from a clip brought in through the Media Pool, do the following:

- 1 Select the clip in the Node Editor.
- 2 In the Inspector, click the Audio tab and select the clip name from the Audio Track drop-down menu.

If more than one MediaIn node exists in the comp, the audio last selected in the Inspector is heard. You can use the Speaker icon in the toolbar to switch between the MediaIn node audio files.
- 3 Right-click the Speaker icon in the toolbar, then choose the MediaIn for the clip you want to hear.

Purging the Audio Cache

The audio and its settings are cached for faster performance. If you change which audio tracks you want to play back in Fusion, or you use the Sound Offset wheel to slip the audio tracks, you need to purge the audio cache. Also, if you return to the Edit, Cut, or Fairlight page and modify the audio levels, you need to purge the audio cache.

To purge the audio cache after any change to the audio playback:

- Click the Purge Audio Cache button in the Inspector.

The audio will be updated when you next playback the composition.

MediaOut Node [MO]



The MediaOut node

NOTE: The MediaOut node is only available in DaVinci Resolve.

MediaOut Node Introduction

Every composition you create in DaVinci Resolve's Fusion page must include a MediaOut node. The MediaOut node sends the final output back to your Timeline on DaVinci Resolve's Edit or Cut page. In most cases, it replaces the Saver node used in Fusion Studio for exporting clips.

The composition output by the Fusion page's MediaOut node is propagated via the Color page's source inputs, with the sole exception that if you've performed transforms or added plugins to that clip in the Edit or Cut page, then the handoff from the Fusion page to the Color page is as follows:



When using Resolve Color Management or ACES, each MediaOut node converts the output image back to the Timeline color space for handoff to the Color page.

NOTE: Additional MediaOut nodes can be added to the Node Editor from the Effects Library. Additional MediaOut nodes are used to pass mattes to the Color page.

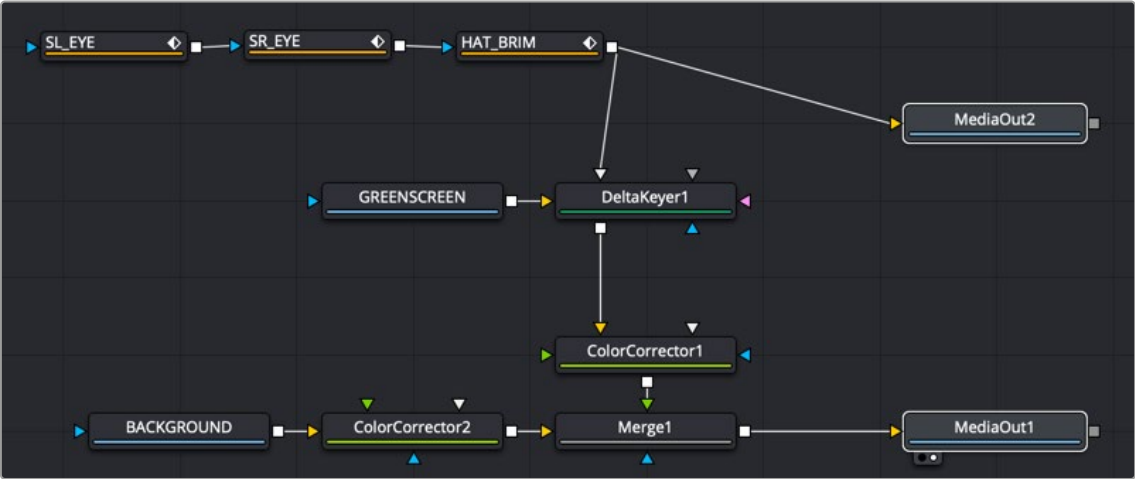
Inputs

The single input on the MediaOut node is where you connect the final composite image you want rendered back into the Edit page.

Input: The orange input is a required input. It accepts any 2D image that you want rendered back into the Edit page.

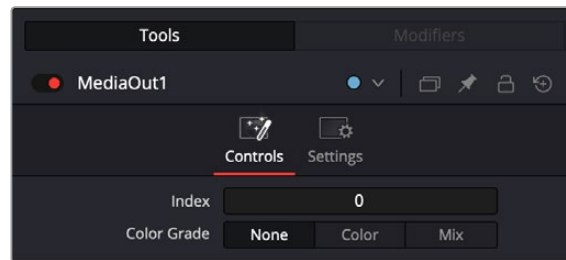
Basic Node Setup

Multiple MediaOut nodes can exist in a single comp. The first MediaOut node always renders the image back to the Edit page Timeline. Additional MediaOut nodes can be used to pass mattes to the Color page.



MediaOut1 node rendering to the Edit page, and MediaOut2 sending mattes to the Color page

Inspector

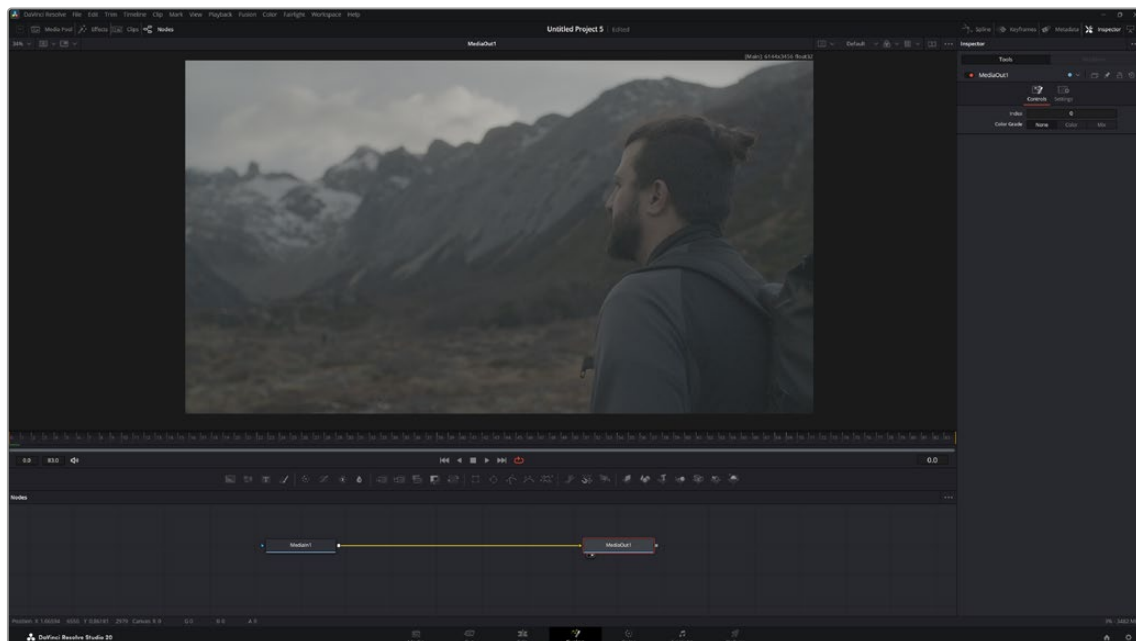


The MediaOut controls

The MediaOut node has various Color Grade options for viewing in the Fusion page.

- **None:** This is the default view, displaying the output of the Fusion composition only.
- **Color:** This displays the output of the Fusion composition, including Color page adjustments at timeline resolution.
- **Mix:** This displays the output of the Fusion composition including Cut and Edit page effects and Color page adjustments at timeline resolution.

NOTE: These options are previews only, to show your current comp in the context of any other effects which may have been added outside Fusion. The actual output of the MediaOut node itself remains unchanged.



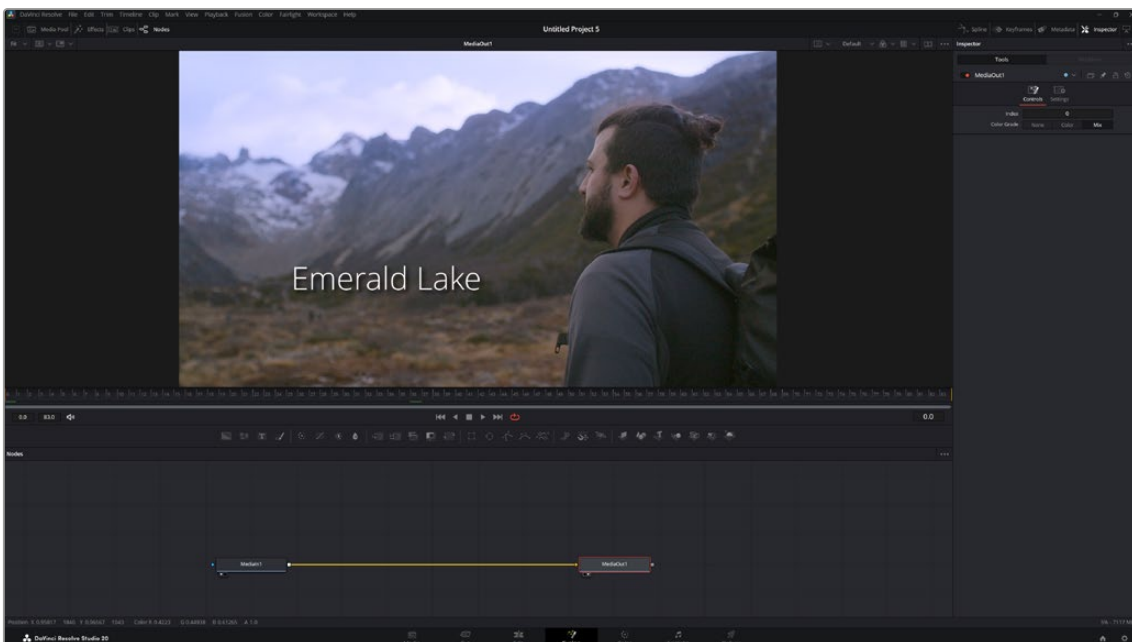
The Color Grade tab set to None

None is displaying the output of the Fusion composition only. Note the display is at the source media/composition resolution.



The Color Grade tab set to Color

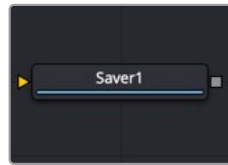
Color is displaying the Fusion composition, including any adjustments applied in the Color page. Note that when the MediaOut view is set to Color, the display will be set to the timeline resolution.



The Color Grade tab set to Mix

Mix is displaying the Fusion composition, including any effects applied in the Cut or Edit page and any adjustments in the Color page. Note that when the MediaOut view is set to Mix, the display will be set to the timeline resolution.

Saver Node [SV]



The Saver node

NOTE: The Saver node in DaVinci Resolve is only used for exporting EXR files.

Saver Node Introduction

The Saver node represents the final composition output from Fusion Studio. It is used to render out movie files or sequential images but can be inserted into a composition at any point to render out intermediate stages of a composition. A composition can contain any number of Saver nodes for rendering different branches of a comp as well as different formats.

The Saver node can also be used to add scratch track audio to your composition, which can be heard during interactive playback.

Inputs

The single input on the Saver node is for the final composition you want to render.

Image Input: The orange input is used to connect the resulting image you want rendered.

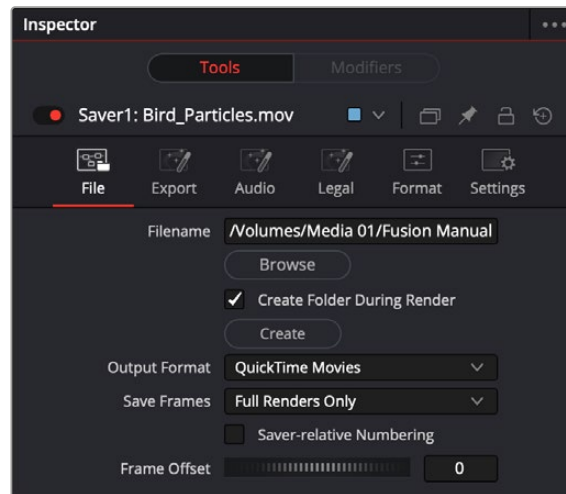
Basic Node Setup

The Saver node is placed at the end of the composition. Multiple Savers can be placed in a comp to render different formats or to render different parts of a composition.



Saver node added to the end of a node tree to render the composition

Inspector



Saver File tab

File Tab

The Saver File tab is used to set the location and output format for the rendered file.

Filename

The Filename dialog is used to select the name and path of the rendered image output. Click on the Browse button to open a file browser and select a location for the output.

Sequence numbering is automatically added to the filename when rendering a sequential image file format. For example, if `c:\renders\image.exr` is entered as the filename and 30 frames of output are rendered, the files are automatically numbered as `image0000.tga`, `image0001.exr`, `image0003.exr`... and so on. Four-digit padding is automatically used for numbers lower than 10000.

You can specify the number of digits to use for padding by explicitly entering the digits into the filename.

For example, `image000000.exr` would apply 6-digit padding to the numeric sequence, `image.001.exr` would use 3-digit padding, and `image1.exr` would use none.

Output Format

This menu is used to select the image format to be saved. Be aware that selecting a new format from this menu does not change the extension used in the filename to match. Modify the filename manually to match the expected extension for that format to avoid a mismatch between name and image format.

Save Frames

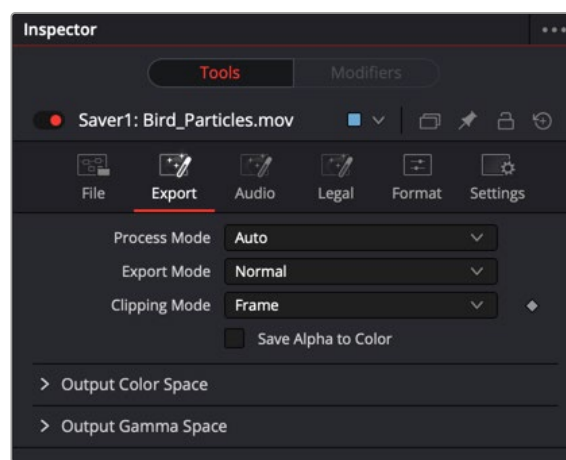
This control selects between two modes of rendering: Full Renders Only or High Quality Interactive.

- **Full Renders Only:** This is the common setting for most situations. Images are saved to disk when a final render is started using the Start Render button in the Time Ruler.
- **High Quality Interactive:** This render mode is designed for real-time rendering when painting and rotoscoping. Fusion saves each frame to disk as it is processed interactively. When used correctly, this feature can eliminate the need to perform a final render after rotoscoping.

NOTE: The High Quality Interactive setting can easily cause confusion when used in conjunction with a node tree that contains spline-animated parameters. If these splines are modified in such a way that frames already saved interactively are changed, the frames already on the disk do not automatically re-render. Either step through each frame again or perform a final render to make certain that the result is correct

Frame Offset

This thumbwheel control can be used to set an explicit start frame for the number sequence applied to the rendered filenames. For example, if Global Start is set to 1 and frames 1-30 are rendered, files are normally numbered 0001-0030. If the Sequence Start Frame is set to 100, the rendered output would be numbered from 100-131.



Saver Export tab

Export Tab

Process Mode

Use this menu to select the Fields Processing mode used by Fusion when saving the images or movie file to disk. The Has Fields checkbox control in the Frame Format preferences determines the default option, and the default height as well. Available options include:

- Full frames
- NTSC fields
- PAL/HD fields
- PAL/HD fields (reversed)
- NTSC fields (reversed).

The two reversed options save fields in the opposite order and thus result in the fields being spatially swapped both in time order and in vertical order as well.

Export Mode

This menu is used to render out the file normally or apply a SMPTE standard 3:2 pulldown to the footage, converting the footage from 24 fps to 30 fps.

Clipping Mode

This menu, sometimes considered source image clipping, defines how the edges of the image should be treated.

- **Frame:** The default Frame setting clips to the parts of the image visible within its visible dimensions. It breaks any infinite-workspace behavior. If the upstream DoD is smaller than the frame, the remaining area in the frame is treated as black/transparent.
- **None:** This setting does not perform any source image clipping at all. This means that any data that would normally be needed outside the upstream DoD is treated as black/transparent. Be aware that this might create humongous images that can consume a considerable amount of disk space. So you should use this option only if really needed.

For more information about ROI, DoD, and Infinite Workspace, see *Chapter 69, "Using Viewers."* in the DaVinci Resolve Reference Manual, or Chapter 7 in the Fusion Reference Manual.

Save Alpha to Color

When enabled, this control causes the Alpha channel to be saved into the color channels as a grayscale image. This completely overwrites any existing color information.

Color Space Type

This menu is used to set the Color Space of the output file so you could, for example, color space convert images from linear to Rec709, etc., thereby delivering linear EXRs, 709 Quicktimes, etc. from a single source. The images are not being converted in the Comp, only in the images saved to disk are converted.

- **Auto:** Passes along any metadata that might be in the rendered image.
- **Space:** Allows the user to set the color space based on the output format.

Curve Type

This menu is used to select a Gamma curve of the rendered file. Once the gamma curve type is set, you can choose to apply the curve for output.

- **Auto:** Passes along any metadata that might be in the incoming image.
- **Space:** Allows the user to set the gamma curve based on the selected file format.
- **Log:** Displays the Log/Lin settings, similar to the Cineon Log node.

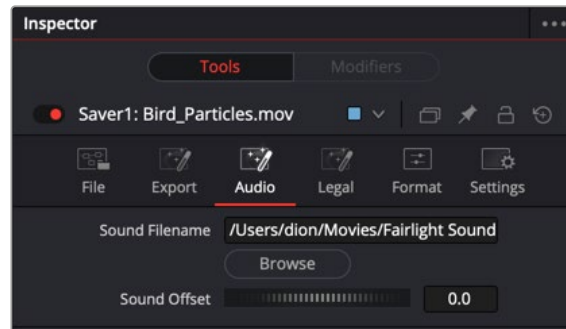
For more detail on the Log settings, see *Chapter 99, "Film Nodes,"* in the DaVinci Resolve Reference Manual, or Chapter 39 in the Fusion Reference Manual.

Apply Curve

Depending on the selected Curve Type or on the Gamma Space found in Auto mode, the associated Gamma Curve is applied, effectively converting from a linear working space.

Audio Tab

NOTE: This tab is only available in Fusion Studio's Saver node.



Saver Audio tab

The audio functionality is included in Fusion Studio for scratch track (aligning effects to audio and clip timing) purposes only. Final renders should almost always be performed without audio. The smallest possible audio files should be used, as Fusion loads the entire audio file into memory for efficient display of the waveform in the Timeline. The audio track is included in the saved image if a Quicktime file format is selected. Fusion currently supports playback of WAV audio.

Source Filename

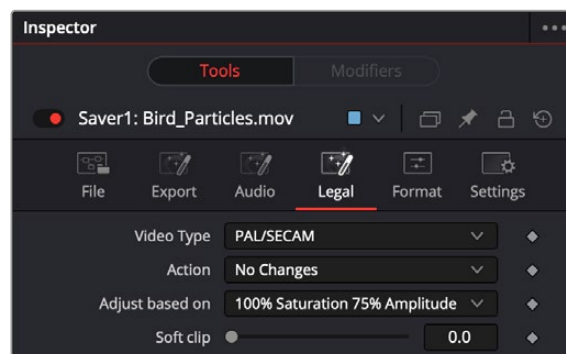
You can enter the file path and name of the audio clip you want to use in the Source Filename field. You can also click the Browse button to open a file browser window and locate the audio scratch track. Select the WAV file of choice, and then in the keyframes panel expand the Saver bar to view the audio waveform. Drag the pointer over the audio wave in the Timeline layout to hear the track.

Sound Offset

Drag the control left or right to slide the Timeline position of the audio clip, relative to other nodes in the Node Editor.

Legal Tab

The Legal tab includes settings for creating “broadcast safe” saturation and video range files for output.



Saver Legal tab

Video Type

Use this menu to select the standard to be used for broadcast legal color correction. NTSC, NHK, or PAL/SECAM can be chosen.

Action

Use this menu to choose how Fusion treats illegal colors in the image.

- **Adjust to Legal:** This causes the images to be saved with legal colors relevant to the Video Type selected.
- **Indicate as Black:** This causes the illegal colors to be displayed as black in the views.
- **Indicate as White:** This causes the illegal colors to be displayed as white in the views.
- **No Changes:** This causes the images to be saved unaffected.

Adjust Based On

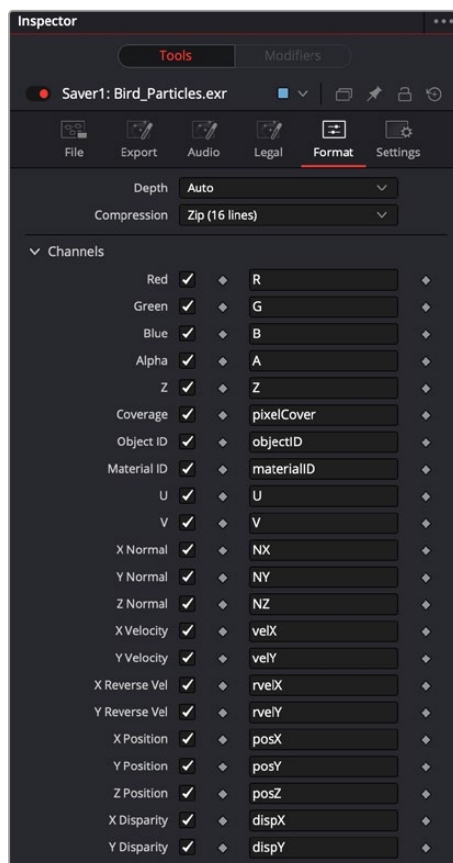
This menu is used to choose whether Fusion makes legal the image to 75% or 100% amplitude. Very few broadcast markets permit 100% amplitude, but for the most part this should be left to 75%.

Soft Clip

The Soft Clip control is used to draw values that are out of range back into the image. This is done by smoothing the conversion curve at the top and bottom of the curve, allowing more values to be represented.

Format Tab

The Format tab contains information, options, and settings specific to the image format being saved. The controls for an EXR sequence is entirely different from the ones displayed when a MOV file is saved.



Saver Format tab

EXR is displayed above for reference.

When the Saver node is set to DPX, it's important to understand the reason for the Bypass Conversion > Data is Linear option. When saving log data into a DPX, and not using the Saver's node's own lin-log conversion (that is, Bypass Conversion is checked), the Data Is Linear option should be off. This indicates whether the reason for checking Bypass Conversion is because the data is linear, or whether it's already log.

If Data Is Linear is enabled, then the DPX is marked in its header as containing linear data. In turn, that means that when the DPX is loaded back into Fusion, or into other apps that evaluate the header, those apps think the data is linear and do not perform any log-lin conversion.

Common Controls

Settings Tab

The Settings tab controls are common to both Loader and Saver nodes, so their descriptions can be found in "The Common Controls" section at the end of this chapter.

Clipping Mode

This menu, sometimes considered source image clipping, defines how the edges of the image should be treated.

- **Frame:** The default Frame setting clips to the parts of the image visible within its visible dimensions. It breaks any infinite-workspace behavior. If the upstream DoD is smaller than the frame, the remaining areas in the frame are treated as black/transparent.
- **None:** This setting does not perform any source image clipping at all. This means that any data that would normally be needed outside the upstream DoD is treated as black/transparent. Be aware that this might create humongous images which can consume a considerable amount of disk space. So you should use this option only if really needed.

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Curve Type

This menu is used to select a Gamma curve of the rendered file. Once the gamma curve type is set, you can choose to apply the curve for output.

- **Auto:** Passes along any metadata that might be in the incoming image.
- **Space:** Allows the user to set the gamma curve based on the selected file format.
- **Log:** Displays the Log/Lin settings, similar to the Cineon Log node.

For more detail on the Log settings, see *Chapter 99, "Film Nodes,"* in the DaVinci Resolve Reference Manual, or Chapter 39 in the Fusion Reference Manual.

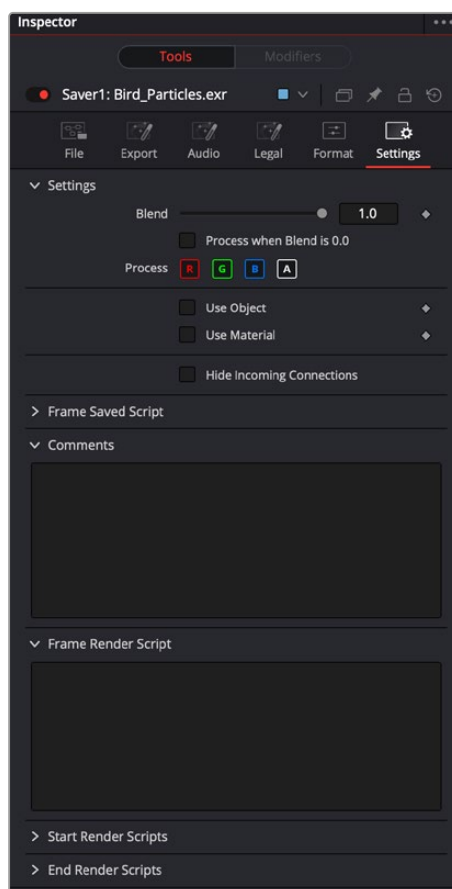
Apply Curve

Depending on the selected Curve Type or on the Gamma Space found in Auto mode, the associated Gamma Curve is applied, effectively converting from a linear working space.

The Common Controls

I/O nodes share a number of identical controls in the Inspector. This section describes controls that are common among I/O nodes.

Inspector



Common Saver settings inspector

Settings Tab

The Settings tab in the Inspector can be found on the Loader, Saver, MediaIn, and MediaOut nodes. The controls are consistent and work the same way for each tool, although some parameters are only available on individual nodes but are covered here.

Blend

The Blend control is used to blend between the tool's original image input and the tool's final modified output image. When the blend value is 0.0, the outgoing image is identical to the incoming image. Normally, this causes the tool to skip processing entirely, copying the input straight to the output.

Process When Blend Is 0.0

The tool is processed even when the input value is zero. This can be useful if processing of this node is scripted to trigger another task, but the value of the node is set to 0.0.

Red/Green/Blue/Alpha Channel Selector

These four buttons are used to limit the effect of the tool to specified color channels. This filter is often applied after the tool has been processed.

For example, if the red button on a Blur tool is deselected, the blur is first applied to the image, and then the red channel from the original input is copied back over the red channel of the result.

There are some exceptions, such as tools for which deselecting these channels causes the tool to skip processing that channel entirely. Tools that do this generally possess a set of identical RGBA buttons on the Controls tab in the tool. In this case, the buttons in the Settings and the Controls tabs are identical.

Apply Mask Inverted

Enabling the Apply Mask Inverted option inverts the complete mask channel for the tool. The mask channel is the combined result of all masks connected to or generated in a node.

Multiply by Mask

Selecting this option causes the RGB values of the masked image to be multiplied by the mask channel's values. This causes all pixels of the image not included in the mask (i.e., set to 0) to become black/transparent.

Use Object/Use Material (Checkboxes)

Some 3D software can render to file formats that support additional channels. Notably, the EXR file format supports Object and Material ID channels, which can be used as a mask for the effect. These checkboxes determine whether the channels are used, if present. The specific Material ID or Object ID affected is chosen using the next set of controls.

Correct Edges

This checkbox appears only when the Use Object or Use Material checkboxes are selected. It toggles the method used to deal with overlapping edges of objects in a multi-object image. When enabled, the Coverage and Background Color channels are used to separate and improve the effect around the edge of the object. If this option disabled (or no Coverage or Background Color channels are available), aliasing may occur on the edge of the mask.

For more information on Coverage and Background Color channels, see *Chapter 78, "Understanding Image Channels,"* in the DaVinci Resolve Reference Manual, or Chapter 18 in the Fusion Reference Manual.

Object ID/Material ID (Sliders)

Use these sliders to select which ID is used to create a mask from the object or material channels of an image. Use the Sample button in the same way as the Color Picker: to grab IDs from the image displayed in the view. The image or sequence must have been rendered from a 3D software package with those channels included.

Hide Incoming Connections

Enabling this checkbox can hide connection lines from incoming nodes, making a node tree appear cleaner and easier to read. When enabled, empty fields for each input on a node are displayed in the Inspector. Dragging a connected node from the node tree into the field hides that incoming connection line as long as the node is not selected in the node tree. When the node is selected in the node tree, the line reappears.

Comments

The Comments field is used to add notes to a tool. Click in the empty field and type the text. When a note is added to a tool, a small red square appears in the lower-left corner of the node when the full tile is displayed, or a small text bubble icon appears on the right when nodes are collapsed. To see the note in the Node Editor, hold the mouse pointer over the node to display the tooltip.

Scripts

Three Scripting fields are available on every tool in Fusion from the Settings tab. They each contain edit boxes used to add scripts that process when the tool is rendering. For more details on scripting nodes, please consult the Fusion scripting documentation.